

GraphRAG on Electronic Health Record: A Knowledge Graph-Enhanced RAG Approach for Healthcare Information Access

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Abstract. The increasing importance of digital health and the need for better health literacy require effective methods to access and understand Electronic Health Records (EHRs). While Large Language Models (LLMs) show promise in this domain, traditional Retrieval-Augmented Generation (RAG) struggles to handle the complex, interconnected nature of clinical data. GraphRAG emerges as a powerful alternative, leveraging knowledge graphs (KGs) to capture semantic relationships within EHRs. This research evaluates the effectiveness of graph expansion in a GraphRAG to enhance information retrieval from FHIR-formatted medical data. We propose a 1-hop expansion approach built upon a lexical search baseline which, while inheriting some limitations of traditional keyword-based retrieval, significantly enhances LLMs' access to comprehensive and diverse contextual information. Our evaluation, using synthetic patient data and a targeted set of questions across five models, reveals that the 1-hop expansion strategy consistently outperforms the baseline in subjective metrics like comprehensiveness and diversity, and frequently in quantitative metrics such as answer and contextual relevancy. These results highlight the potential of our proposal to enhance LLM response when querying intricate medical data.

Keywords: GraphRAG · RAG · Large Language Models · Electronic Health Records.

1 Introduction

Digital health has gained increasing global importance, with initiatives aimed at democratizing access to health information. Programs such as *Healthy People 2030* [21], the introduction of APIs based on the Fast Healthcare Interoperability Resources (FHIR) standard [3], and the *21st Century Cures Act* in the United States [1] highlight the critical need to eliminate health disparities, promote equity, and improve the usability and understanding of health data for all patients. Health literacy encompasses not only access to information but also the ability to find, understand, and effectively use health data [23].

However, the complexity of technical formats and the use of specialized medical jargon significantly limit the accessibility of health information. This is particularly true for patients with low health literacy or language barriers [23]. Consequently, initiatives designed to improve access to electronic health records (EHRs) and health data often fall short of their potential to enhance patient health outcomes. In this landscape, large language models (LLMs) [28] are emerging as transformative tools to bridge these accessibility gaps. Recent studies indicate that LLMs can interpret medical data and generate explanations in natural language, making health information more accessible and understandable [25, 19, 26, 12, 27, 31, 24, 29, 8]. This capability is especially relevant for reducing health disparities and increasing social impact. Despite this potential, traditional retrieval-augmented generation (RAG) approaches encounter challenges when dealing with the intricate interrelationships within clinical data.

GraphRAG offers a significant advancement over traditional RAG by integrating knowledge graphs (KGs) and graph-based techniques like indexing, retrieval, and generation [22]. Unlike traditional RAG, which often struggles with the complex, interconnected nature of healthcare data such as FHIR resources [25], GraphRAG excels at capturing intricate relationships and dependencies. This facilitates a more holistic understanding of patient information, reducing redundancy and enhancing the relevance and contextual richness of generated responses. Specifically within healthcare, approaches like MedGraphRAG demonstrate GraphRAG’s ability to ground LLM responses in external medical knowledge, improving reliability and providing evidence-based health information [29].

In this paper, we employ hop expansion to overcome the limitations of lexical search involving EHRs in FHIR format. This approach is crucial for capturing interconnected information to deliver more comprehensive, detailed, and diverse responses compared to traditional methods. Specifically, we propose and evaluate a 1-hop graph expansion strategy on top of a lexical search baseline to enhance the retrieval of semantically related information from EHR KGs. To this end, we compare the proposed version against the baseline across different quantitative and subjective metrics, aiming to identify the strengths and weaknesses of each approach. This comparative analysis is conducted using synthetic FHIR data representing five distinct patients and a targeted set of eight questions (3 questions from a reference paper [26] and 5 newly proposed questions) for a more comprehensive evaluation, assessed across five state-of-the-art LLMs. We also propose a broad evaluation incorporating subjective and quantitative metrics, alongside latency and token cost analysis, to understand the viability of GraphRAG as a dialogue system. Ultimately, we aim to provide data-driven insights into the nuances of each approach.

2 Background

The application of RAG in healthcare has been extensively explored, with several studies highlighting its potential and limitations. For instance, Ge et al. (2023) utilized RAG to develop a specialized chat application for liver diseases, reporting

favorable outcomes but noting limitations related to the datasets employed [12]. Veen et al. (2024) applied LLMs to the summarization of medical data, achieving performance levels comparable to or exceeding those of medical professionals, though they acknowledged the persistent risk of hallucinations [27]. Additionally, Zaretsky et al. (2024) focused on simplifying hospital discharge summaries for patients, improving readability but encountering challenges in maintaining accuracy and completeness [31]. Saba et al. (2024) and Elgedawy et al. (2024) further explored the use of RAG in electronic medical documents. While Saba et al. (2024) concentrated on summarization [24], Elgedawy et al. (2024) focused on information extraction, both leveraging similarity search methodologies to enhance the utility of medical data [8].

Graph-based approaches, such as GraphRAG, have emerged as powerful tools for enhancing LLM performance in healthcare. Wu et al. (2024) introduced MedGraphRAG, a novel graph-based method designed for the medical domain [29]. MedGraphRAG improves LLM responses by grounding them in evidence and providing explanations of official medical terms. This approach constructs triple graphs that link health data to authoritative medical articles and specialized dictionaries, ensuring reliability and applicability. By utilizing the U-Retrieval strategy, MedGraphRAG optimizes information retrieval through a hierarchical structure of medical tags, balancing efficiency with comprehensive context. Evaluations using medical question benchmarks and clinical assessments demonstrated MedGraphRAG’s + superiority over traditional methods and domain-specific LLMs, establishing a new state-of-the-art in evidence-based healthcare responses. Similarly, Schiffman (2024) presented a proof of concept for employing GraphRAG with FHIR data as an alternative to traditional RAG [25]. The author highlights the suitability of graphs for managing the interconnected nature of FHIR data, facilitating links between resources and offering a more holistic and detailed view of the clinical context. This approach also considers the temporal dimension of medical events, such as analyzing treatment efficacy over time, enabling more detailed and context-aware responses in complex healthcare scenarios.

Beyond healthcare, GraphRAG has been applied to broader domains, demonstrating its versatility and effectiveness. Edge et al. (2024) pioneered GraphRAG for the global summarization of large datasets using KGs derived from LLMs [7]. Departing from traditional graph index retrieval structures, their study emphasizes the inherent modularity of graphs and the capability of community detection algorithms to partition them into related node modules. LLM-generated summaries for these communities comprehensively cover the graph index and input documents. Using a map-reduce approach, each community summary independently addresses queries, which are then consolidated into a global response. Evaluations using questions from real datasets revealed that global methods outperform traditional RAG in comprehensiveness and diversity, as assessed using LLM evaluation techniques [9, 18]. GraphRAG, particularly with community summaries at intermediate and lower levels, demonstrates superior performance

in understanding broad questions and diverse topics. However, this comes with higher token costs.

Several studies have explored the use of graphs to enhance data quality in LLM contexts, employing various strategies to enrich context and improve retrieval. On this basis, the k-hops strategy has been widely used to add relevant contextual information to the target node, as evidenced by several studies [13, 14, 4, 17, 10, 2, 5]. Conversely, Guo et al. (2023) employ Depth-First Search in specific pipeline stages [13], while Dehghan et al. (2024) adopt Beam Search as a search technique [6]. These approaches highlight the flexibility and adaptability of graph-based methods in enhancing LLM performance across diverse applications, further solidifying their role as a critical tool in the evolution of RAG systems.

LLMs have demonstrated significant potential in processing structured healthcare data, particularly when integrated with the FHIR standard. Schmiedmayer et al. (2024) explored the use of LLMs with structured FHIR data to develop a chatbot integrated into a mobile application [26]. This chatbot employs a RAG using a lexical search approach to enhance patient access to and understanding of health information, focusing on democratizing access to health data and improving health literacy. Their study serves as a foundational reference for this paper, which utilizes the same dataset to evaluate solution quality, emphasizing the democratization of health information access. Similarly, KMS Healthcare Innovation Labs (2023) developed a product using RAG with similarity search to extract relevant information from electronic medical documents in FHIR format [19]. Their work leveraged LLMs to assess the quality and accuracy of responses across approximately 200 model-generated questions, achieving an 86.3% accuracy rate. This underscores the credibility and feasibility of their approach for practical healthcare applications.

3 Proposal

Our research utilizes lexical search as the foundational information retrieval method for querying FHIR-formatted EHRs. This approach, established as a standard baseline in works such as Schmiedmayer et al. (2024) applied specifically within the EHR FHIR format, leverages the natural language understanding capabilities of LLMs to identify relevant FHIR resources directly from textual queries [26]. Its demonstrated effectiveness in democratizing access to health information makes it a critical starting point for our investigation into more advanced retrieval strategies.

Recent works in knowledge graph-augmented retrieval have demonstrated the value of graph traversal techniques, with k-hop expansion strategies showing particular promise for enhancing context in question-answering systems. These strategies, which extend the retrieval process beyond the initially matched nodes to their neighbors within a KG, are seen as a powerful way to capture additional meaningful information. As detailed in the Section 2, a number of studies have shown that by incorporating related entities and their connections (e.g.,

through Breadth-First Search, Depth-First Search, Beam Search, 1-hop, 2-hop, or multi-hop expansions), the context provided to LLMs becomes semantically richer, leading to more informed and nuanced answers. However, the application of these methods to FHIR-formatted EHRs remains unexplored, despite the natural alignment between FHIR’s reference architecture and graph-based representations.

Our proposal introduces a 1-hop expansion strategy to capture related information missed during lexical search within the KG. This technique extends the search to the immediate neighbors of the initially retrieved node. The 1-hop approach is chosen for its efficiency, particularly crucial in large KGs where methods like multi-hop, adaptative, BFS and DFS can become computationally expensive, potentially introducing more irrelevant information. Leveraging the connections of FHIR resources in our KG, this focused expansion aims to give LLMs a more complete understanding of patient medical information by including semantically related information from adjacent nodes.

To evaluate the effectiveness of our proposed approach, we utilized a curated set of predefined questions applied in five synthetic patient data. While we included three questions (Q1, Q3, and Q4) from the baseline study [26], we also introduced five novel questions. These additional questions were designed to address a broader range of clinically relevant scenarios, covering aspects such as allergies, care plans, medical billing, procedures, and immunization history, which were not explicitly evaluated in the baseline study.

Finally, to measure the performance of our proposed 1-hop graph expansion strategy against the lexical search baseline, we introduced a combination of both quantitative and subjective metrics to evaluate the proposal. The quantitative evaluation metrics include: Answer Relevancy, Contextual Relevancy, and Hallucination, drawing inspiration from state-of-the-art RAG evaluation frameworks [9, 18]. These metrics provide an automated and detailed assessment of the generated responses. Additionally, we conducted subjective evaluations using metrics such as Comprehensiveness, Diversity, Empowerment, and Directness, adapted from recent literature on GraphRAG papers [7, 32, 15]. These subjective assessments offer a more nuanced understanding of the overall quality and utility of the LLM responses. To ensure the generalizability and robustness of our findings, all experiments were conducted using different cutting-edge LLMs, including both closed-source and open-source models.

4 Methodology

Our methodology follows the pipeline illustrated in Figure 1. The figure outlines both the indexing phase, where the patient KG is constructed from source documents, and the query phase, detailing the steps executed when a user question is processed by the GraphRAG system.

This study utilized synthetic medical data for five patients, extracted from the reference paper [26]. The data were generated using *Synthea*³, a widely used

³ <https://synthetichealth.github.io/synthea/>

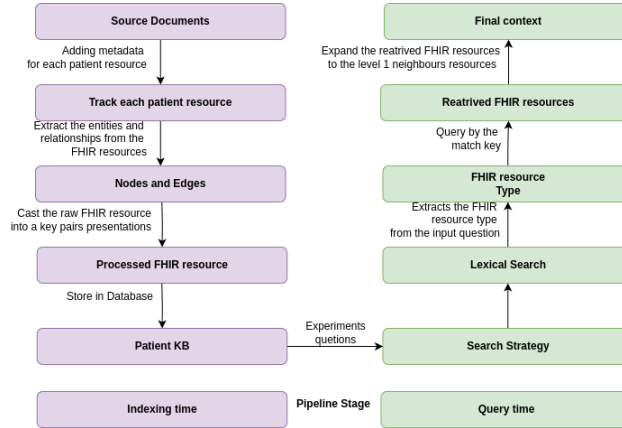


Fig. 1. The pipeline processes FHIR bundle files as source documents. The left side illustrates the indexing phase, detailing the steps to construct the patient KG. The right side depicts the query phase, outlining the steps executed when a question is input into the GraphRAG system.

open-source tool for creating realistic synthetic healthcare data, and were sourced from the publicly available dataset. Table 1 provides a summary of the demographic and key medical characteristics of these five selected patients, including their age, gender, primary conditions, documented allergies, and current medications. This overview illustrated the diversity of patient profiles within the dataset, encompassing a range of ages and complex medical histories. However, the simplified nature of synthetic data necessitates cautious interpretation of results, acknowledging potential limitations in generalizability to real-world EHRs.

Table 2 further details the distribution of different types of FHIR resources across these five patients. It presents the counts for major resource categories such as Conditions, Observations, Encounters, Medications, Procedures, Diagnostic Reports, Claims, and Explanation of Bills, along with the total number of FHIR resources for each patient. This table underscored the richness and interconnectedness of the medical data, highlighting the need for advanced retrieval mechanisms to effectively navigate and extract relevant information [11]. Furthermore, the selection of the FHIR standard was justified by its increasing prevalence and standardization across the healthcare industry, promoting interoperability and data exchange [3]. FHIR’s well-defined structure and constraints provide a robust foundation for building advanced information retrieval systems.

A fundamental step in our GraphRAG methodology was the construction of a KG. This process was crucial for explicitly representing the semantic relationships inherent in the FHIR data, thereby enabling the provision of enriched context to LLMs. These relationships allowed for a more comprehensive retrieval of relevant information, overcoming the limitations often associated with traditional similarity-based search methods in RAG pipelines [22, 30]. In this study, the KG was developed following an enhanced approach proposed by Schifman

Table 1. Summary of the five selected patient data from the work of Schmiedmayer et al. (2024) [26].

Patient	Age	Gender	Conditions	Allergies	Medications
Allen332	82	M	Chronic sinusitis Hypertension Served in armed forces Received higher education Body mass index 30+- obesity Prediabetes Anemia Opioid abuse Atrial ... Carcinoma in situ of prostate Chronic intractable ... partner abuse Stress Alzheimer's disease	Animal dander Penicillin V Peanut	Galantamine 4 MG Warfarin Sodium 5 MG doxycycline hyclate 100 MG DOCEtaxel 20 MG/ML Leuprolide Acetate 30 MG/ML lisinopril 10 MG Verapamil Hydrochloride 40 MG Digoxin 0.125 MG
Beatris270	8	F	Aortic valve stenosis Perennial allergic rhinitis Atopic dermatitis	Latex Bee venom Mold House dust mite Animal dander Grass pollen Tree pollen Aspirin	Fexofenadine hydrochloride 30 MG Epinephrine 1 MG/ML
Edythe31	49	F	Body mass index 30+- obesity Received higher education Prediabetes Anemia Victim of intimate partner abuse Cardiac Arrest History of cardiac arrest	-	Jolivet 28 Day Pack
Jacklyn830	72	F	Essential hypertension Miscarriage in first trimester Ischemic heart disease Chronic kidney disease stage 3 Proteinuria due to type 2 diabetes mellitus Social isolation Sprain (morphologic abnormality)	-	Nitroglycerin 0.4 MG/ACTUAT Simvastatin 20 MG Clopidogrel 75 MG Metoprolol succinate 100 MG Acetaminophen 325 MG Hydrochlorothiazide 25 MG
Milton509	26	M	Hypertension Hypoxemia Stress	-	amLODIPine 2.5 MG

Table 2. Main Resource Type Counts and Total FHIR Resources by Patient

Patient	Condition	Observation	Encounter	MedicationR.	Procedure	DiagnosticR.	Claim	ExplanationOfB.	Total
Allen322	63	252	83	191	63	124	273	82	1235
Beatris270	15	233	31	6	18	33	37	37	500
Edythe31	28	132	35	16	34	55	51	35	442
Jacklyn830	69	1488	56	47	217	243	103	103	2405
Milton509	28	705	19	11	92	104	30	19	1129

(2024) [25]. The primary objective was to transform FHIR data into a graph structure, where FHIR resources were represented as nodes and the references between them as edges.

The key pipeline steps is detailed as follows: First, metadata was added to distinguish and track individual patients within the KG. Second, a script was used to extract FHIR resources and their references, forming the nodes and edges of the graph, respectively. Third, the raw FHIR resources were converted into a key-value pair text representation. Finally, this textual representation, along with the established graph structure, was stored in the KG. This transformation into a structured textual format was designed to optimize token usage for LLMs and had been shown to enhance the quality of their responses [19, 16].

In the query stage, our information retrieval strategy focused on lexical search. This approach leveraged the LLM's inherent understanding of language to infer the relevant FHIR resource type based on the input question [26]. Once the resource type was inferred, it was used to query the KG. The retrieval process

incorporated a 1-hop graph expansion mechanism to capture related information as described in the Section 3.

The resulting subgraph from the retrieval stage was represented in the Adjacency List format. While Guo et al. (2023) noted potential challenges in using textual graph representations with LLMs, they also highlighted the effectiveness of the Adjacency List and Graph Modeling Language (GML) formats [13]. We selected the Adjacency List format due to its lower verbosity compared to GML, which was crucial for managing the LLM’s context window. To further address context window limitations, if the retrieved graph exceeded the model’s capacity, a truncation step was applied. This involved removing text exceeding the limit to ensure the remaining context was usable. This approach was chosen to enable a fair comparison across different LLMs by avoiding the use of summarization techniques that could introduce unwanted variability in content processing.

For the evaluation of this project, we utilized a set of eight predefined questions described in Table 3. Three of these questions (Q1, Q3, and Q4) were adopted from Schmiedmayer et al. (2024) work, as they were designed to assess the LLM’s ability to answer questions based on patient data without requiring medical recommendations [26]. To broaden the evaluation scope, we included five novel questions (Q2, Q5, Q6, Q7, and Q8) to assess aspects such as allergies, care plans, medical billing, procedures, and immunization history, which were not covered in the referenced study. This expansion of the question set ensures a more comprehensive assessment of the retrieval strategies across diverse and meaningful healthcare contexts.

Table 3. Questions evaluated for each patient in this study.

ID	Question
Q1 [26]	What are my current medications, and how should I take them?
Q2	What are my documented allergies, and how severe are they?
Q3 [26]	Can you summarize my current medical conditions?
Q4 [26]	What are my recent laboratory values, what do they mean, and how can I improve them?
Q5	Can you summarize my care plan history?
Q6	Can you provide a breakdown of my medical bills?
Q7	What procedures have I undergone recently, and what were the outcomes?
Q8	Can you summarize my immunization history?

To validate our proposed methodology, these eight evaluation questions were applied to the five patient datasets. The evaluation process involved the use of a diverse set of LLMs, including closed-source models: Gemini-1.5-pro⁴, GPT-4o-2024-08-06⁵, and Claude-v3-opus⁶; and open-source models: Llama-3-70b-instruct-awq⁷ and DeepSeek-R1-Distill-Llama-70B⁸. A fixed system prompt was

⁴ <https://cloud.google.com/vertex-ai/generative-ai/docs/models/gemini/1-5-pro>

⁵ <https://platform.openai.com/docs/models/gpt-4o>

⁶ <https://www.anthropic.com/news/claude-3-family>

⁷ <https://huggingface.co/casperhansen/llama-3-70b-instruct-awq>

⁸ <https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Llama-70B>

used across all experiments to ensure consistency and isolate the impact of the retrieval strategies. We implemented and compared two lexical search strategies: (i) the L0 with no graph expansion, serving as a baseline inspired by Schmiedmayer et al. (2024) [26]; and (ii) the L1 with 1-hop graph expansion, representing our proposed enhancement.

For evaluation, we employed a combination of quantitative and subjective metrics, drawing inspiration from established RAG evaluation frameworks [9, 18], either for question-level and patient-level analysis. The quantitative metrics included: Answer Relevancy measured the extent to which the LLM’s response directly answered the input question, Contextual Relevancy assessed the relevance of the retrieved context to the input question, indicating how well the retrieved information supported the generation of accurate answers, and Hallucination identified instances where the LLM generated factually incorrect information by comparing its output against the provided context, which provides automated evaluations with detailed reasoning for each score.

In addition, we conducted subjective evaluations to assess the overall quality of the generated responses, included: Comprehensiveness evaluated the extent to which the answer covered all relevant aspects of the question, Diversity assessed the variety of perspectives and insights provided in the answer, Empowerment measured how well the answer enabled the reader to make informed judgments, and Directness assessed the specificity and clarity with which the answer addressed the question. These subjective metrics were used for head-to-head comparisons between the L0 and L1 strategies per patient, LLM model, and question. For each search strategy, an Overall Winner was selected based on a holistic assessment of performance across these subjective criteria. Tied comparisons were excluded from the final analysis of subjective metrics [7, 32, 15].

Furthermore, to understand the practical implications of our approach, we performed a technical feasibility study [8], to analyze the total number of tokens required for generating responses within our GraphRAG architecture. This analysis provided insights into the system’s computational demands and its potential scalability for real-world applications.

In total, we evaluated 400 LLM responses (5 patients x 8 questions x 5 LLMs x 2 search strategies). Complete source code, results, and implementation details are available in our public GitHub repository⁹. The experimental setup utilized several key libraries, including Langchain with OpenAI and VertexAI APIs for managing interactions with LLMs, and DeepEval[18] for evaluating the performance. A Neo4j graph database was employed for storing and querying the underlying graph structures. All experiments were conducted on a workstation equipped with an AMD Ryzen 7 5700X processor and 64GB of RAM.

5 Experimental Results

This section presents the experimental results, beginning with the findings from the subjective evaluation. Table 4 presents question-level results, while Table 5

⁹ <https://github.com/baptvit/graphrag-on-fhir>

details the results across the individual patients. Both tables report the evaluation for the five subjective metrics: Comprehensiveness, Diversity, Empowerment, Directness, and the Overall Winner preference. For each metric, the highest value between the two strategies is indicated in bold, highlighting the preferred approach based on the subjective assessment.

Table 4. Comparison of Lexical Search Strategies by Question in choice proportions

Question	Comprehensiveness		Diversity		Empowerment		Directness		Overall Winner	
	L0	L1	L0	L1	L0	L1	L0	L1	L0	L1
Q1	0.200	0.800	0.200	0.800	0.240	0.760	0.560	0.440	0.280	0.720
Q2	0.330	0.670	0.330	0.670	0.330	0.670	0.460	0.540	0.330	0.670
Q3	0.120	0.880	0.400	0.600	0.440	0.560	0.800	0.200	0.360	0.640
Q4	0.320	0.680	0.280	0.720	0.240	0.760	0.200	0.800	0.240	0.760
Q5	0.080	0.920	0.120	0.880	0.240	0.760	0.800	0.200	0.160	0.840
Q6	0.240	0.760	0.560	0.440	0.480	0.520	0.480	0.520	0.320	0.680
Q7	0.200	0.800	0.320	0.680	0.520	0.480	0.800	0.200	0.320	0.680
Q8	0.160	0.840	0.440	0.560	0.680	0.320	0.830	0.170	0.320	0.680

Table 5. Comparison of Lexical Search Strategies by Patient in choice proportions

Patient	Comprehensiveness		Diversity		Empowerment		Directness		Overall Winner	
	L0	L1	L0	L1	L0	L1	L0	L1	L0	L1
Allen322	0.300	0.700	0.425	0.575	0.500	0.500	0.564	0.436	0.375	0.625
Beatris270	0.075	0.925	0.300	0.700	0.325	0.675	0.575	0.425	0.225	0.775
Edythe31	0.231	0.769	0.333	0.667	0.462	0.538	0.692	0.308	0.308	0.692
Jacklyn830	0.275	0.725	0.325	0.675	0.375	0.625	0.700	0.300	0.375	0.625
Milton509	0.150	0.850	0.275	0.725	0.325	0.675	0.550	0.450	0.175	0.825

The results consistently indicate that the L1 strategy outperforms the baseline L0 across the majority of subjective metrics, both when evaluated by question and by patient. This improvement is particularly evident in Comprehensiveness, Diversity, and Empowerment, where L1 consistently achieves higher scores. For instance, when addressing the question of current medications (Q1 [26]), L1 demonstrated substantially higher choice proportions in comprehensiveness and diversity compared to L0, suggesting that incorporating 1-hop graph expansion provides a richer and more nuanced understanding of the information being retrieved.

Similarly, for the question about summarizing medical conditions (Q3 [26]), L1 achieved significantly higher choice proportions in comprehensiveness, diversity, and empowerment. This trend holds across the patient-level analysis in Table 5, where patients like Beatris270, who have a diverse set of observations (Table 2), show marked improvements in comprehensiveness and diversity with L1. The higher Overall Winner percentages for L1 further support its effectiveness in delivering more satisfactory answers. However, the Directness metric

presents a more varied picture, with L0 sometimes outperforming L1, indicating a potential trade-off where the expanded context might slightly reduce the directness of the response, a phenomenon also noted by Edge et al. (2024)[7]. For example, in Q1, L0 achieved a higher directness score compared to L1.

The quantitative evaluation, presented in Tables 6 for question-level and 7 for patient-level, compares L0 and L1 across three key metrics: Answer Relevancy Score (ARS), Contextual Relevancy Score (CRS), and Hallucination Score (HALS). All scores are reported as mean $\pm$ standard deviation (mean $\pm$ std) over multiple evaluations, with bold values denoting superior performance for each metric.

Table 6. Comparison of Lexical Search Strategies by Question

Question	ARS		CRS		HALS	
	L0	L1	L0	L1	L0	L1
Q1	0.660 $\pm$ 0.360	0.790$\pm$0.220	0.400 $\pm$ 0.220	0.470$\pm$0.250	0.170 $\pm$ 0.390	0.160$\pm$0.370
Q2	0.980$\pm$0.020	0.770 $\pm$ 0.420	0.930$\pm$0.090	0.870 $\pm$ 0.290	0.000$\pm$0.000	0.080 $\pm$ 0.290
Q3	0.910$\pm$0.090	0.890 $\pm$ 0.160	0.670 $\pm$ 0.220	0.810$\pm$0.170	0.050$\pm$0.200	0.430 $\pm$ 0.510
Q4	0.830 $\pm$ 0.340	0.990$\pm$0.040	0.620 $\pm$ 0.260	0.620$\pm$0.180	0.590$\pm$0.500	0.710 $\pm$ 0.470
Q5	0.950 $\pm$ 0.210	0.990$\pm$0.020	0.900$\pm$0.190	0.760 $\pm$ 0.260	0.040 $\pm$ 0.210	0.040 $\pm$ 0.210
Q6	0.990 $\pm$ 0.040	1.000$\pm$0.010	0.640 $\pm$ 0.180	0.740$\pm$0.170	0.500$\pm$0.510	0.620 $\pm$ 0.500
Q7	0.950 $\pm$ 0.210	0.950$\pm$0.080	0.840 $\pm$ 0.190	0.880$\pm$0.180	0.130$\pm$0.340	0.380 $\pm$ 0.500
Q8	0.990$\pm$0.040	0.920 $\pm$ 0.120	0.930$\pm$0.150	0.880 $\pm$ 0.160	0.000$\pm$0.000	0.120 $\pm$ 0.340

Table 7. Comparison of Lexical Search Strategies by Patient

Patient	ARS		CRS		HALS	
	L0	L1	L0	L1	L0	L1
Allen322	0.930 $\pm$ 0.180	0.970$\pm$0.050	0.700 $\pm$ 0.270	0.790$\pm$0.230	0.180$\pm$0.380	0.300 $\pm$ 0.470
Beatris270	0.880 $\pm$ 0.300	0.940$\pm$0.130	0.850$\pm$0.200	0.770 $\pm$ 0.230	0.100$\pm$0.310	0.200 $\pm$ 0.400
Edythe31	0.890$\pm$0.250	0.880 $\pm$ 0.220	0.650 $\pm$ 0.290	0.710$\pm$0.280	0.230$\pm$0.430	0.390 $\pm$ 0.490
Jacklyn830	0.960$\pm$0.090	0.920 $\pm$ 0.200	0.730 $\pm$ 0.220	0.750$\pm$0.260	0.260 $\pm$ 0.450	0.170$\pm$0.390
Milton509	0.870$\pm$0.280	0.860 $\pm$ 0.240	0.680$\pm$0.300	0.640 $\pm$ 0.260	0.290$\pm$0.460	0.400 $\pm$ 0.500

The quantitative metrics provide further insights into the performance of the search strategies. For most questions, L1 demonstrates higher or comparable ARS and CRS compared to L0. This suggests that expanding the search context not only improves the subjective quality of the answers but also enhances their relevance and contextual grounding. For example, when asked about recent laboratory values (Q4 [26]), L1 achieved a notably higher ARS compared to L0, indicating a better alignment with the user’s information need. This pattern is also observed at the patient level; for instance, Allen322, who has a significant number of Observation and DiagnosticReport resources described in Table 2, shows a higher ARS with L1 compared to L0.

However, the variations in HALS highlight a critical challenge in GraphRAG systems. While expanding the search context through graph relationships can improve contextual relevance, it also introduces a higher risk of hallucinations; this is further explored in long-context degradation problems [20]. This trade-off between contextual richness and answer relevancy is a key consideration in designing effective GraphRAG architectures. The results suggest that the choice of search strategy should be carefully tailored to the specific application requirements, balancing the need for comprehensive information retrieval with the imperative of factual accuracy. For applications requiring high factual accuracy and contextual relevance, Lexical Search strategies L0 might be preferred in scenarios where the risk of hallucination needs to be minimized, especially for questions like Q2 concerning allergies where precision is paramount. However, for tasks where diversity and richness of information are more critical, L1 may be appropriate, provided that mechanisms are in place to mitigate hallucination risks.

To further understand the practical implications of these search strategies, we analyzed their computational efficiency. The data reveals that the adoption of L1 results in a noticeable increase in both processing time and token usage compared to the L0. The mean total processing time for L1 was 33.38 seconds, compared to 23.57 seconds for L0. Similarly, the mean total token usage for L1 was 62,730 tokens, significantly higher than the 16,216 tokens for L0. This increase in resource consumption is a direct consequence of the larger context that needs to be processed when incorporating graph expansion. The higher median and maximum values for both time and tokens in L1 further emphasize this increased computational burden. For instance, the median token usage for L1 is 35,251, more than five times that of L0 with 6,630. This suggests that while L1 often leads to improved answer quality, it comes at a cost of higher computational resources, which might be a limiting factor in certain applications or with large-scale KGs, potentially exacerbating the long-context degradation issues [20]. The trade-off between enhanced performance and computational efficiency highlights the need for careful optimization of graph expansion techniques in real-world GraphRAG deployments.

Our experimental results demonstrate that L1 generally enhances the quality of lexical search results, particularly in terms of comprehensiveness, diversity, and empowerment. It often leads to higher or comparable answer and contextual relevancy to baseline. However, this improvement is accompanied by a potential increase in hallucination rates and a significant rise in computational cost, as reflected in processing time and token usage. The choice between L0 and L1 strategies should therefore be guided by the specific requirements of the application, balancing the need for enriched contextual information with constraints on computational resources and the criticality of factual accuracy. Further research into optimizing graph expansion techniques, mitigating the risk of hallucinations, and addressing potential long-context degradation issues is crucial for advancing the effectiveness of GraphRAG systems in complex domains like healthcare.

6 Conclusions and Future Works

In this research, we addressed the increasing need for efficient and comprehensive information retrieval from EHRs by proposing and investigating the effectiveness of 1-hop graph expansion within a GraphRAG framework utilizing data in the FHIR format. Our primary objective was to determine if incorporating graph-based relationships during the retrieval process could significantly enhance the quality of LLM responses when querying complex medical data, evaluating its impact on subjective metrics like comprehensiveness, diversity, empowerment, and directness, as well as quantitative metrics such as answer relevancy, contextual relevance, and hallucination.

The experimental evaluation of lexical search strategies, with and without 1-hop graph expansion, revealed that L1 consistently outperformed L0 in subjective evaluations, demonstrating substantial improvements in Comprehensiveness, Diversity, and Empowerment across various clinical questions and patient datasets. This highlights the advantage of incorporating contextual graph relationships to provide richer and more insightful answers, even if it occasionally led to a slight reduction in Directness. Furthermore, L1 generally showed comparable or better Answer and Contextual Relevancy Scores. However, this enhancement came with a notable increase in computational cost, as evidenced by higher processing times and token usage, and presented a potential for increased hallucination rates in certain scenarios.

Future work should investigate strategies to optimize graph expansion techniques to mitigate increased computational resources and hallucination potential. This involves exploring graph modularity combined with intermediate summarization to decrease token consumption while preserving low hallucination likelihood and ensuring comprehensive, diverse, and interconnected answers from EHRs. Furthermore, expanding research scope should incorporate larger, diverse patient datasets and broader query arrays to evaluate the graph expansion strategy. Alternative retrieval methods, like vector-based retrieval with clinical embeddings (e.g., BioBERT), alongside sophisticated context management techniques (e.g., hierarchical graph prompting, dynamic context window allocation and summarization) may mitigate bias introduced by graph context truncation.

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